Project A: More Modes Than You Think

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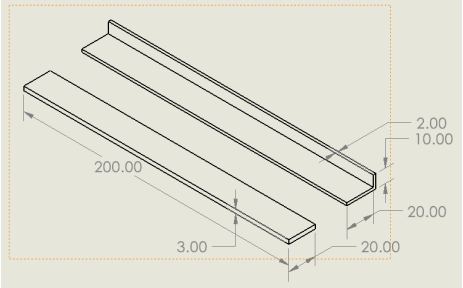
AE 264: Vibrations, Spring 2026

ABSTRACT

This project examines the modal behavior of an aircraft wing (idealised as a cantilever beam) using both analytical models and finite element analysis (FEA). Natural frequencies for bending, torsional, and axial modes are first predicted using Euler–Bernoulli theory and related formulations, then compared with results from a 3D ANSYS model. The comparison highlights discrepancies arising from neglected effects such as Poisson's effect, shear deformation, rotary inertia, and warping. Additionally, bending–torsion coupling is investigated by comparing a rectangular cross-section with an L-shaped section, showing that symmetry leads to decoupled modes while shear-centre eccentricity introduces coupled behavior. The study demonstrates the limitations of simplified analytical models and the importance of 3D effects in accurately capturing structural dynamics.

Given beam properties:

- Material: Aluminium 6061-T6
- $E = 70\,\text{GPa}$, $\rho = 2700\,\text{kg/m}^3$, $\nu = 0.33$
- Dimensions of beam with rectangular section: $\ell = 200\,\text{mm}$, $b = 20\,\text{mm}$, $h = 3\,\text{mm}$



PHASE 1: ANALYTICAL PREDICTIONS

(i) Cantilever Frequency Equation

For an Euler–Bernoulli (EB) beam free vibration, the transverse displacement satisfies $EI\,W^{IV} = \omega^2 m\,W$, whose general solution is given by the equation below:

$$W_n(x) = A\cosh\beta_n x + B\sinh\beta_n x + C\cos\beta_n x + D\sin\beta_n x,$$

where the parameter β_n is defined as:

$$\beta_n^4 = \frac{m\,\omega_n^2}{EI}.$$

Applying the fixed-end boundary conditions, $W(0) = 0$ and $W'(0) = 0$ yields the relations $C = -A$ and $D = -B$. Substituting these back into our general solution gives:

$$W(x) = A(\cosh\beta x - \cos\beta x) + B(\sinh\beta x - \sin\beta x).$$

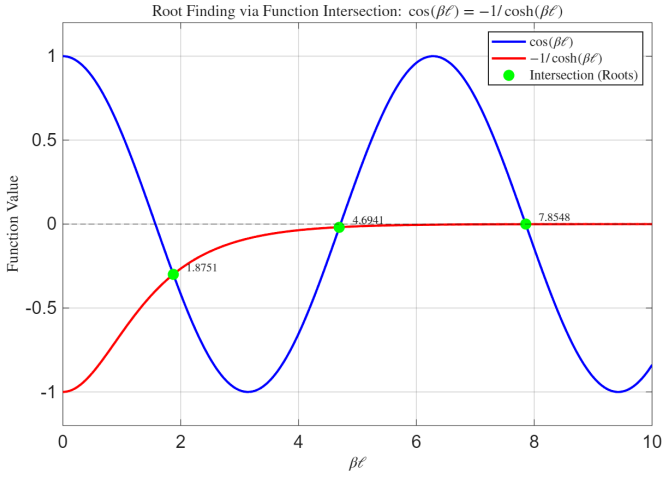
At the free end ($x = \ell$), the condition for zero bending moment is $W''(\ell) = 0$ and for zero shear force is $W'''(\ell) = 0$. This produces the following linear system:

$$\begin{pmatrix} \cosh\beta\ell + \cos\beta\ell & \sinh\beta\ell + \sin\beta\ell \\ \sinh\beta\ell - \sin\beta\ell & \cosh\beta\ell + \cos\beta\ell \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \mathbf{0}.$$

For a non-trivial solution to exist, the determinant of the coefficient matrix must be set to zero. Utilizing the hyperbolic and trigonometric identities $\cosh^2\theta - \sinh^2\theta = 1$ and $\cos^2\theta + \sin^2\theta = 1$, we arrive at the characteristic equation:

$$\boxed{\cosh\beta\ell\,\cos\beta\ell + 1 = 0.}$$

Solving this transcendental equation numerically, we find the first three roots to be $\beta_1\ell = \mathbf{1.8751}$, $\beta_2\ell = \mathbf{4.6941}$, and $\beta_3\ell = \mathbf{7.8548}$.



(ii) Vertical Bending Frequencies (Weak-Axis)

First, we calculate the section properties for bending about the weak axis:

$$I_z = \frac{bh^3}{12} = \frac{0.02 \times 0.003^3}{12} = 4.50 \times 10^{-11}\,\mathrm{m}^4, \quad m = \rho bh = 2700 \times 0.02 \times 0.003 = 0.162\,\mathrm{kg/m}$$

The natural frequencies are then determined using the formula:

$$f_n = \frac{(\beta_n \ell)^2}{2\pi}\sqrt{\frac{EI_z}{m\,\ell^4}}$$

Where the structural constant evaluates to:

$$\sqrt{\frac{EI_z}{m\,\ell^4}} = \sqrt{\frac{70 \times 10^9 \times 4.50 \times 10^{-11}}{0.162 \times (0.2)^4}} = 110.24\,\mathrm{s}^{-1}$$

Mode	$\beta_n \ell$	f_n^{VB} (Hz)	Ratio f_n/f_1
1	1.8751	**61.6**	1.00
2	4.6941	**386.0**	6.27
3	7.8548	**1080.9**	17.55

Table 1: Predicted vertical (weak-axis) bending frequencies derived from Euler-Bernoulli theory.

(iii) Predicted Frequencies of All Mode Types

While Euler-Bernoulli theory focuses on simple bending, a real 3D solid beam can also undergo lateral bending, torsional, and axial vibrations. We must estimate these to understand the full modal landscape.

Lateral bending. The relevant second moment of area is calculated about the strong axis:

$$I_y = \frac{hb^3}{12} = \frac{0.003 \times 0.02^3}{12} = 2.00 \times 10^{-9}\,\mathrm{m}^4$$

The same EB formula applies, substituting I_y:

$$f_{L1} = \frac{1.8751^2}{2\pi} \times \sqrt{\frac{EI_y}{m\ell^4}} = \frac{1.8751^2}{2\pi} \times 735.1 = \mathbf{410.6}\,\mathrm{Hz}$$

Torsion. The fundamental torsional frequency is estimated as:

$$f_{T1} = \frac{1}{4\ell}\sqrt{\frac{GJ}{\rho\, I_p}} = \frac{1}{4 \times 0.2}\sqrt{\frac{26.32 \times 10^9 \times 1.80 \times 10^{-10}}{2700 \times 2.045 \times 10^{-9}}} = \mathbf{1158}\,\mathrm{Hz}$$

Longitudinal (axial). The fundamental axial mode depends on the material wave speed:

$$f_{A1} = \frac{1}{4\ell}\sqrt{\frac{E}{\rho}} = \frac{1}{4 \times 0.2}\sqrt{\frac{70 \times 10^9}{2700}} = \mathbf{6365}\,\mathrm{Hz}$$

Mode type	Predicted f_1 (Hz)
1st vertical bending	**61.6**
2nd vertical bending	**386.0**
1st lateral bending	**410.6**
1st torsional	**1158**
1st axial	**6365**

Table 2: Predicted frequencies for the 3D cantilever beam.

PHASE 2: FEA DISCOVERY (ANSYS 3D SOLID)

Model Setup

The cantilever beam is modeled in ANSYS using a 3D solid cubic mesh. One end face is fixed constraining all six degrees of freedom. The material used is Al 6061-T6 with the given analytical parameters. A modal analysis is then executed to extract the first 12 natural frequencies and their corresponding mode shapes.

Results: Rectangular Cross-Section

Mode	Mode type	FEA f (Hz)	EB prediction (Hz)	% error
1	Vertical Bending (VB1)	62.326	61.69	1.02%
2	Vertical Bending (VB2)	390.23	386.60	0.93%
3	Lateral Bending (LB1)	409.33	411.26	-0.47%
4	Vertical Bending (VB3)	1092.4	1082.49	0.91%
5	Torsion (T1)	1151.8	1157.78	-0.52%
6	Vertical Bending (VB4)	2141.1	2121.25	0.93%
7	Lateral Bending (LB2)	2453.7	2577.33	-5.04%
8	Torsion (T2)	3474.6	3473.34	0.04%
9	Vertical Bending (VB5)	3540.3	3506.57	0.95%
10	Vertical Bending (VB6)	5289.6	5238.21	0.97%
11	Torsion (T3)	5855.4	5788.90	1.14%
12	Axial	6379.2	6364.69	0.23%

Table 3: First 12 modes from ANSYS modal analysis compared with theoretical predictions (rectangular cross-section). The % error indicates the degree to which EB theory underestimates the FEA frequencies.

The corresponding visual representations of these mode shapes are illustrated in Figure 1.

Results: L-Shaped Cross-Section

To investigate the effects of asymmetry, the cross-section is modified to an L-shape while preserving the cross-sectional area ($bh \approx 60\,\text{mm}^2$). The L-shape has a total overall height of 10 mm, a total overall width of 22 mm, and a uniform thickness of 2 mm for both the vertical and horizontal legs. The same modal analysis procedure is repeated.

The visual representations of these mode shapes are illustrated in Figure 2.

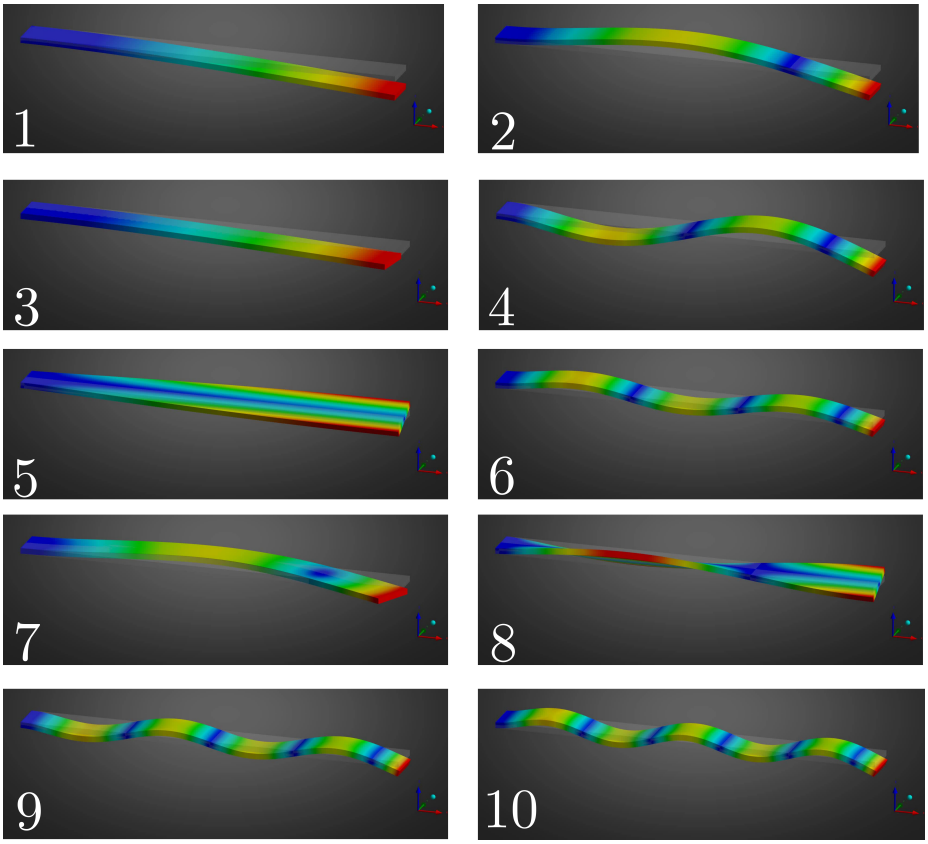


Figure 1: First 10 mode shapes of the rectangular beam obtained via ANSYS modal analysis.

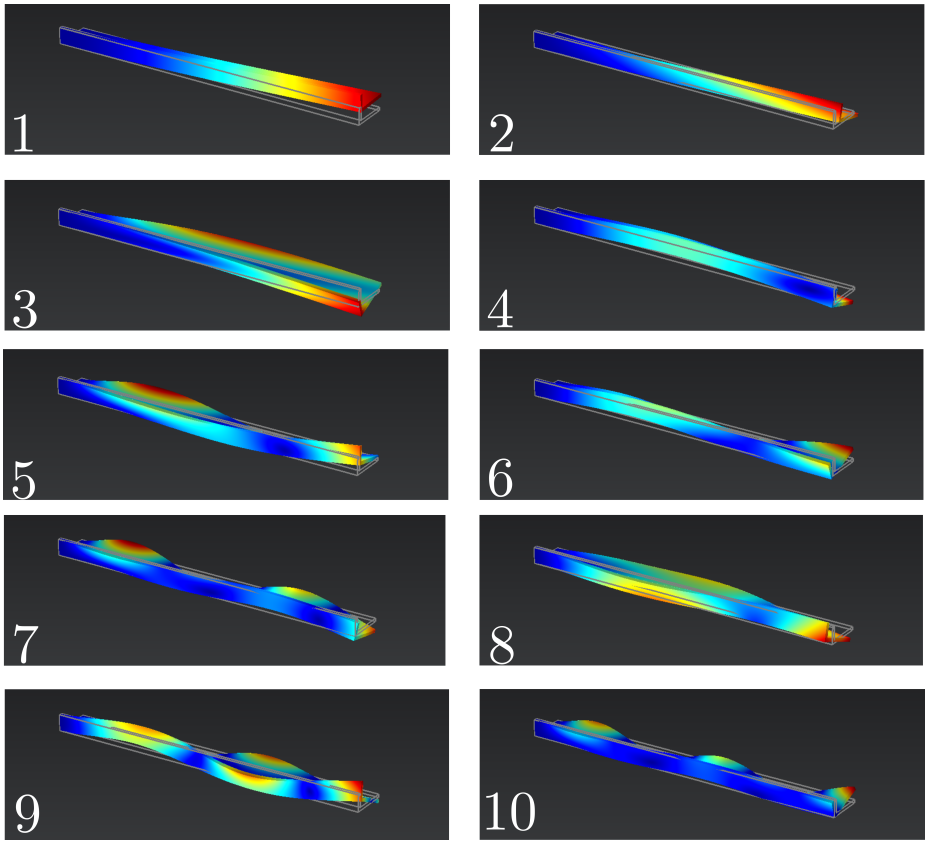


Figure 2: First 10 mode shapes of the beam with 'L' shaped cross-section obtained via COMSOL modal analysis.

PHASE 3: EXPLAIN

(i) Vertical Bending Modes: Theoretical Overestimation vs. Reality

According to classical theory, natural frequencies are governed by the relation $f \propto \sqrt{K/M}$, where K is the effective stiffness and M is the effective mass. One-dimensional Euler-Bernoulli (EB) beam theory inherently assumes that plane cross-sections remain perfectly flat and perpendicular to the neutral axis during flexure. In reality, a 3D beam is subjected to transverse shear deformation (which adds structural flexibility, reducing K) and rotary inertia (which introduces rotational kinetic energy, increasing M). Consequently, EB theory generally overestimates natural frequencies.

However, Table 3 reveals that for the first few vertical bending modes, the FEA results are slightly *higher* than the EB predictions (62.33 Hz vs. 61.69 Hz). This underestimation by the EB model is attributed to 3D root constraints. Bending naturally induces transverse normal strains due to the Poisson effect, causing the cross-section to curl (anticlastic curvature). In the 3D FEA model, the fixed boundary strictly prevents this warping at the root. Denying the material its natural, lowest-energy deformation path induces a complex multiaxial stress state that locally stiffens the beam. For a vertically thin beam ($h = 3$ mm), this stiffening effect outweighs the negligible shear and rotary effects, resulting in a net increase in the actual natural frequency.

(ii) Lateral Bending Modes: The Dominance of Shear Deformation

For lateral bending, the principal axis changes, giving an effective depth in the plane of bending equal to $b = 20$ mm. The EB theory predicts a fundamental lateral frequency of 411.26 Hz, whereas the FEA yields 409.33 Hz.

In this orientation, the EB theory overestimates the frequency. This occurs due to the fundamental assumption of EB theory that plane cross-sections remain perfectly perpendicular to the neutral axis, effectively assuming infinite shear rigidity. In reality, transverse shear deformation contributes to the total compliance. Because the dimension parallel to the plane of bending (20 mm) is significantly larger than the vertical thickness (3 mm), the shear strain energy becomes a non-negligible fraction of the total deformation energy. By ignoring this added flexibility, the EB mathematical model is artificially stiff, resulting in an overestimated natural frequency.

Furthermore, Table 3 demonstrates that this overestimation error increases rapidly for higher modes (from -0.47% in LB1 to -5.04% in LB2). This rapid divergence is heavily dependent on the modal wavelength. As the mode number increases, the spatial distance between inflection points (the wavelength) becomes drastically shorter. The influence of both transverse shear deformation and rotary inertia scales inversely with the square of the wavelength. As the wavelength approaches the cross-sectional depth of the beam, the EB assumptions violently break down, making the analytical model progressively much stiffer than the physical 3D beam.

(iii) Torsional Modes: Edge Effects and Warping Restraint

The analytical model for uniform cantilever torsion yields a fundamental frequency of 1157.78 Hz, utilizing the thin-rectangle approximation for the torsion constant, $J \approx bh^3/3$. The corresponding FEA result is slightly lower at 1151.8 Hz.

Table 3 shows mode-dependent trend in the torsional error: it begins negative (-0.52% for T1) but shifts entirely positive for higher modes ($+1.14\%$ for T3).

(iv) Decoupling in the Rectangular Cross-Section

An examination of the rectangular beam's mode shapes obtained from the 3D FEA confirms that the flexural and torsional behaviors are strictly independent. As illustrated in the modal visualizations, the rectangular beam exhibits either pure vertical bending, pure lateral bending, or pure torsion. There is no overlapping or mixing of these motions. A single mode does not show both bending and twisting simultaneously.

(v) Bending-Torsion Coupling in the L-Shaped Cross-Section

Conversely, the mode shapes of the L-shaped cross-section exhibit pronounced bending-torsion coupling. In these modes, flexural displacement is inherently accompanied by twisting simultaneously.

This stark difference in behavior between the two cross-sections is governed by the relative locations of the cross-section's centroid (center of mass) and its shear centre (the theoretical elastic axis through which transverse forces must act to produce pure bending without twist).

In the rectangular beam, double symmetry dictates that the centroid and shear center occupy the exact same geometric location. Because transverse inertial forces generated during flexure act strictly through the centroid, and the centroid lies directly on the shear center, there is no moment arm (the eccentricity, e, is zero). Thus, no localized torque is generated, and the modes remain entirely independent.

However, the L-profile is asymmetric. Its shear center resides near the intersection of the two orthogonal flanges, while its centroid is located in the open space between them. Consequently, there is a physical offset, or eccentricity ($e \neq 0$), between the mass axis and the elastic axis.

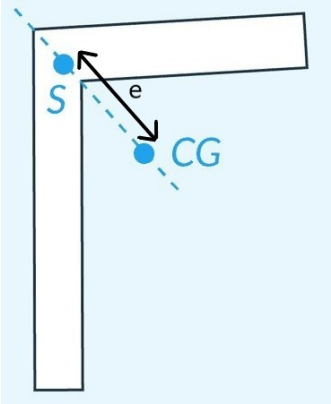


Figure 3: L-shaped cross-section illustrating the physical offset (eccentricity) between the shear center (S) and the centroid (CG).

Mathematically, this eccentricity forces the general equations of motion to entangle. For simultaneous transverse displacement $v(x,t)$ and twist $\phi(x,t)$, the dynamic inertial forces are governed by a non-diagonal mass matrix:

$$\begin{bmatrix} m & me \\ me & I_0 \end{bmatrix} \begin{bmatrix} \ddot{v} \\ \ddot{\phi} \end{bmatrix}$$

where m is the mass per unit length and I_0 is the polar mass moment of inertia about the shear centre. Because the off-diagonal coupling terms (me) are non-zero, it is physically impossible to isolate the motions. Any flexural acceleration $\ddot{v}$ generates an inertial torque ($me\ddot{v}$) that forces the beam to twist. Simultaneously, any angular acceleration $\ddot{\phi}$ generates a transverse inertial force ($me\ddot{\phi}$) that forces the beam to bend. Therefore, unlike the symmetric rectangle, every natural mode of the L-shaped beam is inherently a coupled hybrid.